EXPERIMENT-5b

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Simulate the Life Cycle Stages for UI Design Using the RAD Model and Develop a Small Interactive Interface Using OpenProj

AIM

The aim of this experiment is to simulate the lifecycle stages of UI design using the Rapid Application Development (RAD) model and to design a small interactive Login and Registration interface. The project schedule is tracked and managed using OpenProj, an open-source project management tool.

Tool Used

- OpenProj – for project scheduling and Gantt chart creation
- Axure RP – for wireframing and interactive prototyping
- Figma – for high-fidelity visual design
- Tool Link: https://sourceforge.net/projects/openproj/

Procedure

Step 1: Requirements Planning

The first step involves identifying the core requirements for the Login and Registration UI. The goal is to understand what the interface must achieve and what interactions users will perform.

1. Gather Requirements:

- Identify key features: Login form, Registration form, form validation, and error messages.
- Define the target user: Any end-user who needs to authenticate into a system.
- Specify interface elements: Username field, Password field, Email field, Submit button, and navigation link between Login and Register screens.

2. Define Use Cases:

- Use Case 1: User logs in with valid credentials and is redirected to the dashboard.
- Use Case 2: User attempts login with invalid credentials and sees an error message.
- Use Case 3: New user registers with a unique username and email and receives a confirmation message.

Output in OpenProj:

- Created a new project titled "Login & Register UI – RAD Model".
- Added tasks: "Gather Requirements" (Duration: 2 days) and "Define Use Cases" (Duration: 2 days).
- Set dependencies and assigned resources to each task.

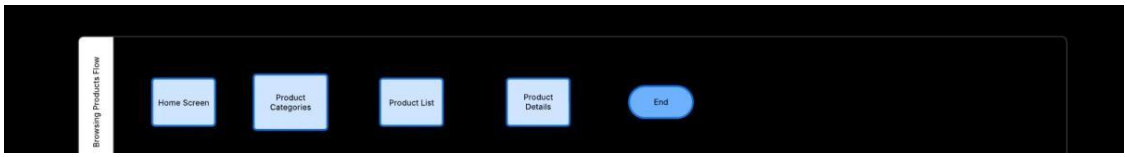

Fig 1: OpenProj – Requirements Planning Tasks

Step 2: User Design

In this step, initial sketches and digital wireframes are created for the Login and Register screens. The wireframes define the layout, input fields, and navigation between screens.

1. Sketch Initial Designs:

- Drew rough paper sketches showing placement of fields and buttons on both screens.
- Identified common elements: page header, form container, input labels, and action buttons.

2. Create Digital Wireframes:

- Login Screen: Username field, Password field, "Login" button, and "Register here" link.
- Register Screen: Username field, Email field, Password field, Confirm Password field, and "Register" button.
- Both screens include input validation indicators and error message placeholders.

Output in OpenProj:

- Added tasks: "Sketch Initial Designs" (Duration: 1 day) and "Create Digital Wireframes" (Duration: 3 days).
- Allocated time and resources; set dependency on Requirements Planning phase.

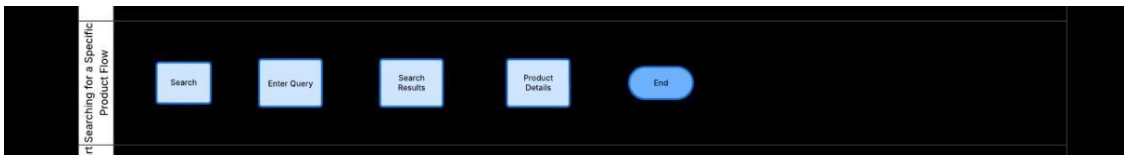

Fig 2: OpenProj – User Design Tasks

Step 3: Rapid Prototyping

The wireframes are converted into interactive prototypes using Axure RP. The prototype simulates actual application behaviour, allowing stakeholders to experience the interface flow before development begins.

1. Develop Prototypes:

- Built clickable prototype in Axure RP with navigation between Login and Register screens.
- Added form validation behaviour: error messages on empty submission, password mismatch alert.
- Simulated successful login redirect to a placeholder dashboard screen.

2. Test Prototypes:

- Shared prototype link with stakeholders via Axure Cloud.
- Collected feedback: users requested larger button sizes and clearer field labels.
- Iterated on design: increased font size, added placeholder text in fields, and improved colour contrast.

Output: Interactive prototypes for "Login" and "Register" screens.

Output in OpenProj:

- Added tasks: "Develop Prototypes" (Duration: 3 days) and "Test Prototypes" (Duration: 2 days).
- Set milestones and dependencies on User Design phase.

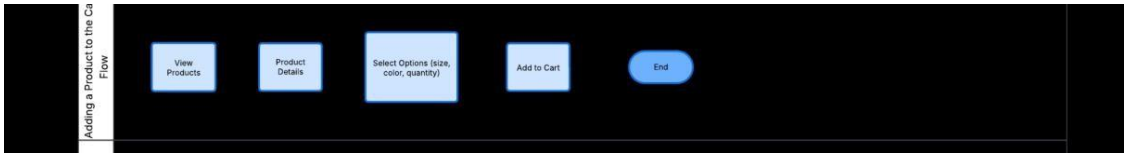

Fig 3: OpenProj – Rapid Prototyping Tasks

Step 4: User Acceptance / Testing

In this step, the completed prototype is reviewed by users and stakeholders. Usability testing is conducted to identify pain points and validate that the interface meets user expectations.

1. Review Prototype:

- Conducted a walkthrough session with stakeholders demonstrating the full Login and Register flow.
- Gathered observations: navigation was intuitive; minor issues with error message visibility were noted.

2. Conduct Usability Testing:

- Five users tested the prototype and were asked to complete tasks: "Login with given credentials" and "Register a new account".
- Documented feedback: all users completed tasks successfully; two users requested a "Show Password" toggle.
- Updated prototype to include password visibility toggle and improved error message styling.

Output: Documented feedback and test results with proposed design improvements.

Output in OpenProj:

- Added tasks: "Review Prototype" (Duration: 2 days) and "Usability Testing" (Duration: 3 days).
- Tracked progress and assigned review resources.

Fig 4: OpenProj – User Acceptance & Testing Tasks

Step 5: Implementation

The final step involves implementing the approved interface design. Based on all collected feedback and approved prototype, the functional interface is developed and integrated with backend services where required.

1. Develop Functional Interface:

- Implemented the Login screen using HTML, CSS, and JavaScript with client-side validation.
- Implemented the Register screen with form validation for email format, password strength, and field matching.
- Applied the approved colour scheme, typography, and spacing from the final prototype.

2. Integrate Backend (if required):

- Connected Login form to a backend authentication API for credential verification.
- Connected Register form to a user database to store new account details securely.
- Implemented session management to redirect authenticated users to the dashboard.

Result

The experiment was successfully completed. The lifecycle stages of UI design using the RAD model were simulated for a Login and Registration interface. Each stage — Requirements Planning, User Design, Rapid Prototyping, User Acceptance Testing, and Implementation — was documented and tracked using OpenProj. The Gantt charts confirm that all tasks were completed within the scheduled durations. The final prototype provides a fully interactive Login and Register interface that meets all specified user requirements.

Conclusion

The RAD model proved to be an effective approach for developing the Login and Registration UI. By using iterative prototyping and continuous user feedback at each stage, the design was progressively refined to meet user expectations. OpenProj provided a clear visual representation of the project timeline, helping manage tasks, resources, and dependencies efficiently. This experiment demonstrated the practical application of project management tools alongside UI design methodologies.